

PRG: Physics-Aware Reward Guidance for Text-Conditioned Human Motion Diffusion

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Project Final Report

1. Introduction

Text-conditioned human motion diffusion models generate naturalistic motion sequences, yet they routinely produce physically infeasible trajectories. Existing methods address this by embedding a full reinforcement-learning-trained physics simulator inside the denoising loop, which is effective but expensive and brittle when the denoised estimate is still far from a plausible trajectory. To resolve this, we will propose a simulator-free gradient-steering framework that enforces physical plausibility through a feasibility reward model. Contrary to the conventional assumption that structural motion fidelity emerges gradually throughout the sampling process, our empirical findings demonstrate that the fundamental kinematic characteristics are established within the initial 10% of denoising steps, at which point the generated motion already aligns visually with the conditioning prompt. Our approach will bypass the computational bottlenecks of iterative physics projection by injecting a learned reward-guidance signal directly into the denoising steps of a frozen Motion Diffusion Model (MDM) backbone.

This reward model will be trained contrastively, learning to differentiate between high-fidelity motion capture data and a diverse set of "physically impossible" generated data. We investigate reward guidance scheduling through two contrasting strategies: a decreasing schedule to prevent over-correction as $\hat{x}_0$ stabilizes, and an increasing schedule that grants exploratory freedom early while enforcing physical feasibility in the final stages. We will benchmark our method on HumanML3D and report FID, R-Precision and physics metrics against the MDM baseline.

2. Dataset, Task, Metrics

2.1. Dataset

We use HumanML3D [23], the standard benchmark for text-conditioned human motion generation, from the official repository: <https://github.com/EricGuo5513/HumanML3D>. Constructed by re-annotating two large motion capture databases (AMASS [4] and HumanAct12 [9]) with natural language description, it consists of 14,616 motion clips and 44,970 annotations with standard train/val/test splits. Each frame is represented as a 263D vector including root motion, joint kinematics, velocities, and foot contact labels. Its scale, diversity, and physics-relevant features make it well-suited for our setting. Its explicit foot contact labels and joint velocity features enable direct computation of our physical plausibility metrics without additional annotation, and its text descriptions serve dual purpose as training supervision and inference prompts.

To train the feasibility reward R_ϕ , we construct a contrastive dataset of physically plausible and implausible motions. Ground-truth sequences from HumanML3D serve as positive examples, while negative samples are synthesized via MDM; specifically, generated motions that exhibit poor physical metrics, such as excessive ground penetration or foot sliding—are classified as negatives. This contrastive framework enables the reward model to effectively distinguish between kinematically fluid motion and physically inconsistent artifacts.

2.2. Task: Text-to-Motion Generation with Dynamic Feasibility

The task is text-conditioned motion synthesis - given a natural language description (e.g., "a person walks forward and then turns left"), generate a physically plausible human motion sequence. Existing methods including our Motion Diffusion Model (MDM)[19] baseline produce kinematically coherent but physically implausible motions, exhibiting foot sliding and dynamically infeasible joint accelerations. Text-to-motion is the ideal testbed as it demands both prompt-alignment and physical realism, a tension that reward-guided diffusion is specifically designed to resolve. HumanML3D annotations supervise text-motion correspondence during training and are reused as conditioning prompts at inference, directly enabling our generation task.

2.3. Metrics

Following the evaluation framework established by PhysDiff [22], we evaluate our model across two dimensions: generative quality and physical plausibility. We use the open-source evaluation code from MDM [19] for standard motion metrics and implement physics-based metrics with a 5mm tolerance to account for geometry approximation.

Generative Quality: We report **FID** to measure the distribution distance between generated and ground-truth motions, and **R-Precision (Top-1)** to assess text-motion alignment. R-Precision is calculated as the fraction of samples where the ground-truth description c_i is correctly retrieved among $K = 32$ candidates: $\frac{1}{N} \sum_i \mathbf{1}[\text{rank}(c_i | \hat{x}_0^{(i)}) = 1]$.

Physical Plausibility: We use the following three metrics to quantify geometric violations, using the sum of these averages to define the total physical error (*Phys-Err*).

- **Ground Penetration (GP):** Measures the total depth of joints passing below the floor plane ($y = 0$). Following the formulation in Fig. X:

$$GP = \sum_{t=1}^T \sum_{j \in J} \max(0, -y_{t,j}) \quad (1)$$

where T is the number of frames, J is the set of joints, and $y_{t,j}$ is the vertical position of joint j at time t .

- **Floating (Float):** Measures the distance between the ground and the lowest body joint when it remains above the floor during expected contact:

$$Float = \sum_{t=1}^T \max(0, y_{min,t} - \epsilon) \quad (2)$$

where $y_{min,t}$ is the lowest joint height at frame t and $\epsilon = 5\text{mm}$ is the allowed tolerance.

- **Foot Skating (Skate):** Quantifies the horizontal drift of feet during contact frames. It is calculated as the average horizontal displacement Δp_{xy} for joints where $y < \epsilon$:

$$Skate = \frac{1}{N_{contact}} \sum_{t,j} \mathbf{1}[y_{t,j} < \epsilon] \cdot \|p_{t+1,j} - p_{t,j}\|_{xy} \quad (3)$$

3. Related Work

Motion Diffusion Models: Modern text-to-motion frameworks have transitioned from VAE-based architectures to large-scale diffusion models. Key works such as the **Motion Diffusion Model (MDM)** [19] introduced direct diffusion in motion space using transformers, while **MotionDiffuse** [20] provided fine-grained control through probabilistic mapping. Subsequent iterations like **MLD** [21] further optimized these processes by operating in efficient latent representations. These models provide the generative backbone that our work augments via external guidance rather than structural replacement.

Physics-Guided Generation: To bridge the gap between kinematic plausibility and physical feasibility, researchers have looked toward physics-in-the-loop approaches. **PhysDiff** [22] represents the state-of-the-art in this domain, embedding a full physics simulator (IsaacGym) and a reinforcement learning-based motion imitation policy within the denoising loop. While PhysDiff significantly reduces foot-sliding and penetration on benchmarks like HumanML3D [23], its reliance on an expensive simulator and RL policy at inference time limits its scalability. We seek similar feasibility gains through a lighter, gradient-based guidance mechanism that bypasses the need for full simulation.

Diffusion Guidance: Steering diffusion sampling via auxiliary gradients was established by **classifier guidance** [10] and further extended by **Diffusion Posterior Sampling (DPS)** [24]. DPS is particularly relevant as it enables measurement models to operate on the predicted clean data estimate $\hat{x}_0$, allowing complex rewards to steer trajectories even when the measurement model is not noise-aware. We adapt this paradigm to motion generation, utilizing the backpropagation-through-the-denoiser approximation to steer the diffusion process toward physically consistent manifolds.

Reward-Guided Diffusion: Scalar rewards have been used extensively to align diffusion models with human preferences in image generation [25, 26] and for trajectory optimization in offline reinforcement learning [27, 28]. These methods demonstrate that specialized reward models can guide sampling toward high-quality regions of the data manifold. We extend this paradigm to the motion domain by introducing a feasibility reward. Drawing on classical robotics, such rewards can be constructed using differentiable analytical tools like the **Recursive Newton-Euler Algorithm (RNEA)** [29] or libraries such as **Pinocchio** [30], which allow for the computation of joint torques directly from kinematics.

Noise-Aware Scheduling: The efficacy of guidance is often tied to the variance-exploding or variance-preserving schedules of the diffusion process [31]. Literature suggests that x_0 estimates are systematically less stable at high noise levels, necessitating time-conditioned modulation of the guidance strength. We adopt this insight by applying reward guidance primarily as the motion structure stabilizes, ensuring that physical constraints do not prematurely suppress generative diversity during the initial exploratory steps of denoising.

4. Approach

4.1. Baselines

We evaluate our method against two primary baselines: **MDM** [19] and **PhysDiff** [22]. In our experiments, we replicate the MDM baseline and perform a comprehensive evaluation using the HumanML3D [23] test set, comprising approximately 4,000 motion sequences. We report all established metrics—including FID, R-Precision, and the suite of physical plausibility scores—on these generated samples. It is important to note that we do not replicate the PhysDiff training process; the unavailability of its source code and the prohibitive computational cost of training an imitation learning policy for every denoising step make full replication infeasible. Instead, we compare against the results reported in the original PhysDiff literature where applicable.

4.2. Proposed Approach

Our key contribution is a **novel application** of gradient-steered diffusion guidance [24] to physical motion plausibility, adapting an existing framework to a new domain without retraining the backbone. Our method consists of two components: a feasibility reward model R_ϕ trained offline, and a guided sampler that injects its gradient at inference. Each design choice is motivated by prior work: we freeze the backbone because [24] shows that pretrained diffusion models can be steered toward arbitrary objectives without retraining; we adopt a contrastive objective because RLHF [17] demonstrates that binary preference data yields well-calibrated rewards.

4.2.1. Feasibility Reward Model

The feasibility reward model R_ϕ is implemented as a transformer-based architecture that utilizes Adaptive Layer Normalization (AdaLN) to incorporate timestep conditioning. Unlike simple concatenation, AdaLN modulates the normalization layers within each transformer block based on an embedding of the timestep t , allowing the network to dynamically adapt its feature extraction to the varying noise levels of the motion estimate $\hat{x}_0^t$. During training, R_ϕ is optimized on contrastive pairs of positive $\hat{x}_0^+$ and negative $\hat{x}_0^-$ samples. The positives $\hat{x}_0^+$ are reaped from real HumanML3D sequences and MDM-generated samples which have top physics metric values. The negatives $\hat{x}_0^-$ are obtained from MDM-generated samples with the worst physics metric values. We train the reward model by minimizing the Bradley-Terry ranking objective:

$$\mathcal{L} = -\mathbb{E} [\log \sigma (R_\phi(\hat{x}_0^+, t) - R_\phi(\hat{x}_0^-, t))] \quad (4)$$

This time-conditioned scalar output provides the necessary flexibility to modulate guidance strength as the motion transitions from high-variance noise to a stabilized structure, enabling an empirical investigation into the optimal scheduling of physical constraints throughout the diffusion process.

4.2.2. Guided Sampler

At *inference*, each denoising step applies two sequential corrections. First, CFG produces a prompt-aligned estimate:

$$\hat{x}_{0,\text{cfg}}^t = \text{MDM}(x_t, t, \emptyset) + w_{\text{cfg}}(\text{MDM}(x_t, t, c) - \text{MDM}(x_t, t, \emptyset))$$

Then, for all the denoising timesteps, a physics correction is injected:

$$\tilde{\mu}_t = \mu_t + \sigma_t^2 \cdot w(t) \cdot \nabla_{\hat{x}_{0,\text{cfg}}^t} R_\phi(\hat{x}_{0,\text{cfg}}^t)$$

where $w(t)$ increases monotonically as $t \rightarrow 0$, concentrating corrections where $\hat{x}_0^t$ is well-formed and the gradient is most reliable. The corrected MDM posterior mean estimate is passed to the standard diffusion update to produce x_{t-1} .

5. Experiments

- **MDM Baseline Evaluation:** We evaluate the pretrained MDM checkpoint on the HumanML3D test set to establish a quantitative benchmark, as detailed in Table 1. Our findings reveal a significant discrepancy between the baseline performance reported in the PhysDiff literature and our own replication results. Specifically, while the original metrics for MDM suggest higher physical stability, our replication demonstrates a notable increase in artifacts such as foot skating (9.634 vs. 1.406). We attribute this divergence to the lack of publicly available source code and non-explicit parameterization of evaluation metrics in prior work, which hinders a direct parity comparison.

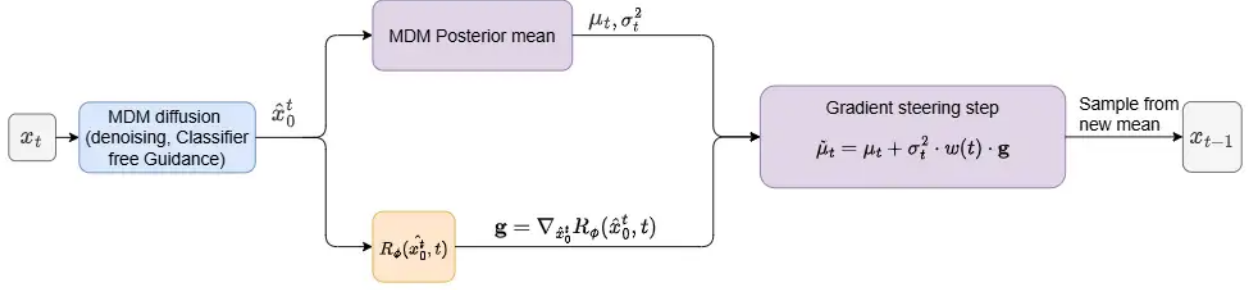


Figure 1: Inference with late-stage reward guidance.

- **Reward Model Training:** Utilizing HumanML3D validation data, we train a base reward model on contrastive positive and negative samples sourced from ground truth and MDM-generated sequences. For our initial experiment, we trained the model for 10 epochs on a subset of 800 samples to verify convergence behavior before scaling to the full dataset. This preliminary model achieved a validation accuracy of 98%, where accuracy is defined by the model’s ability to correctly assign a higher reward to the positive (ground truth) sequence than the negative (MDM-generated) sequence in a pairwise comparison. The loss successfully converged during this phase, confirming that physical plausibility can be captured even with limited supervised data. Following this verification, we will begin training with the complete dataset and perform an extensive architectural and optimization hyperparameter sweep (Table 2).
- **Integrated Reward-Guided Diffusion:** We perform reward-guided sampling on the HumanML3D test data, investigating various time-scheduling strategies (increasing and decreasing) for the guidance weight $w(t)$ to optimize the balance between physical feasibility and generative quality.
 - **Key Observation:** Our findings challenge the assumption of incremental motion refinement; instead, we observe a rapid convergence of kinematic structure during the initial 10% of sampling. As demonstrated in this example ([walking forward trajectory](#)), the core semantic and structural features are solidified—resolving from noise into a recognizable gait between timesteps $t = 49$ and $t = 45$ —far sooner than the final refinements of physical plausibility.

Method	FID↓	R-P↑	Sk↓	Pen↓	Fl↓	Ph↓
MDM[22]	.544	.611	1.41	11.3	18.9	31.6
MDM (<i>rep</i>)	.497	.709	9.63	3.80	17.6	31.0
PhysDiff[22]	.433	.631	.512	.998	2.60	4.11
PRG (Ours)	0.55	0.702	9.84	4.78	12.98	27.6

Table 1: Results on HumanML3D.

Hyperparam.	Search Space
Channels	64, 128 , 256
DiT Blocks	2, 3, 4 , 6
LR / Drop	0.0001, 0.0003, 0.001
Attn. Heads	2 , 4, 8
FFN mult.	1×, 2 ×, 4×
Batch Size	32, 64, 128

Table 2: Sweep Config.

6. Code Overview

Here, we describe important sections of the code that we developed and are relevant to understanding the literature as a whole.

The **AdaLNBlock** class in `model.py` (lines 21-52, `/prg/prg_code/reward_model/model.py`) implements an Adaptive Layer Normalization block for the DiT, where it injects scale/shift dynamically to the normalized activations based on the timestep embedding. The **FeasibilityRewardModel** class in the same file (lines 55-116) implements the core reward model. It uses the AdaLNBlock to implement a DiT model that return scalar scores for denoised motion samples, based on the denoising timestep.

The **reward_cond_fn** function in `reward_cond_fn.py` (lines 87-134, `/prg/prg_code/reward_model/reward_cond_fn.py`) computes the gradient of a reward model for a denoised motion sample at a particular timestep, and returns the scaled gradient that will be used in updating the diffusion posterior mean for physics feasibility steering.

The **select_hard_negatives** function in `score_motions.py` (lines 53-58, `/prg/prg_code/score_motions.py`) scores all the motion trajectories generated by the MDM. By ranking them according to their scores, it retrieves the bottom 25th

percentile of the trajectories and returns them as the hard negative samples.

7. Timeline, Human Work, and AI Use

Week	Milestone	Owner
1–2	Data & Model Training (Mid-report): Collect synthetic/adversarial motion data. Train R_ϕ and submit preliminary results.	Narayanan, Karthik
3	Evaluation: Sweep K and γ ; calculate physics metrics (foot-skating, torque violations), run late-stage gating ablations, and validate gradient flow to $\hat{x}_0$.	Narayanan, Rahul
4	Finalization: Produce qualitative figures, perform ablation analysis, and complete final write-up and presentation.	Karthik, Rahul

Table 3: Project timeline and milestone distribution.

7.1. Human work vs AI

- **Human Work:** All initial code development, including the core MDM replication and reward model training loops were performed by the human team. All conceptual analysis and experimental design are strictly human-led.
- **AI Assistance:** Gemini/GPT-4/Claude were utilized for code refinement, boilerplate utility functions (e.g., standardizing physics metrics and LaTeX formatting), and technical rephrasing for clarity.
- **Verification:** To ensure trustworthiness, all AI-assisted code was unit-tested against ground-truth motions and manually audited for mathematical parity with established literature. AI-generated text was reviewed to prevent technical hallucinations or misrepresentations of data.

8. Research Log

8.1. Inspiration

We were initially inspired by the use of classifier guidance to steer the input towards prompt adherence and wanted to implement something similar for physical plausibility. Based on literature search and brainstorming, we that an approach based on the RLHF objective function would be better suited for physics feasibility guidance than classifier guidance, and we went ahead with training a reward model for checking physics plausibility. The paired-preference dataset we construct for training our reward model and the Bradley-Terry ranking objective for model optimization come from this inspiration (see Section 4.2.1 for more details)

8.2. Dataset Creation

For training a reward model, we require both physically plausible (positive) and implausible (negative) motion sequence samples. Initially, we planned to reap the physically plausible samples from the benchmark, the HumanML3D [23] dataset, and for the negative samples, we planned to use the baseline MDM to synthetically generate bad samples, based on a preset threshold for the physics metrics computed on these samples (see Section 2.1).

But when we actually generated the samples from the MDM, we found that a lot of them demonstrated physics metric values that would bring them under the same class as physically plausible positive samples. Based on this observation, we reconfigured our framework for dataset creation: we sampled 5 different trajectories for each prompt from the HumanML3D train set and then took the best and worst among them as positive and negative samples, respectively. Even in the ground-truth motion samples reaped from the HumanML3D benchmark, our assumption that all of them would be positive turned out to be false. Some noisy ground-truth motion samples turned out to have physics metrics values worse than the MDM generated ones. To reap only the worthy motions, we used a five-grouping technique similar to the one implemented in GRPO, and only label those ground-truth samples as positive whose physics metric values are better than the best MDM sample.

8.3. Algorithm

For the 50-step denoising MDM model, we initially performed an analysis how the denoised motion samples at each timestep fare in terms of physics plausibility. We realized that in the span of denoising timesteps 49 to 45, the trajectory is somewhat aligned with the prompt to the naked eye. We concluded that the bulk of the denoising occurs in the

initial 10% of the denoising timesteps (or 5 timesteps in this case). Based on this conclusion, we hypothesized that physics feasibility steering with the reward model should suffice for just the first 5 denoising timesteps. Since the motion pattern is established in these initial timesteps, we feared that the reward model based gradient steering might lead the model away from prompt adherence. But the results contradicted our hypothesis, we found that steering at all denoising timesteps produced the most optimal results.

Along the same lines, we also considered the tradeoff between prompt adherence and physical plausibility, and wanted to experiment with toning down the gradient update to ensure balance in both. For this purpose, we normalized the reward gradient and updated the diffusion posterior mean with only the unit vector in the direction of the reward gradient. The results were not promising, and we decided to stick with the original unnormalized value of the reward gradient for the posterior mean update. In fact, this worsened the physics metrics: we think the magnitude of the reward gradient is important for successful steering.

Interestingly, our initial plan was to apply gradient steering only to the final denoising timestep (just use final step data for reward model training). We quickly realized that the denoised motion samples already become physically infeasible by the time they reach the final timestep. So, we opted to train a reward model conditioned on timestamp “t” of the denoising process and use it to steer every time step. We believe that this has enabled us to obtain such a meaningful reduction in physics metric errors. Our approach is to assign the same class (positive or negative) to all denoised motion samples from the intermediate timesteps as the final denoised motion sample.

8.4. Understanding Baselines

We spent a good amount of time understanding the results of the baseline models, especially MDM [19]. Physdiff [22], though state-of-the-art for human motion generation, did not correlate with our own results. Physdiff neither released an open-source code base for us to replicate, nor did they detail the implementations of the evaluation metrics they used to report their results. To create a more believable/reproducible baseline, we ourselves ran MDM for 50 denoising steps and obtained the results, which we report in Table 1 along with our own results.

9. Conclusion

We present PRG, a simulator-free framework for enforcing physical plausibility in text-conditioned human motion diffusion via gradient steering from a learned feasibility reward model. On HumanML3D, PRG reduces total physical error by 11% over the MDM baseline (31.0 $\rightarrow$ 27.6), with a 26.3% improvement in the Floating metric specifically, all without an embedded physics simulator or RL policy at inference — the key computational cost of the state-of-the-art PhysDiff. Our experiments also surface two non-obvious findings: kinematic structure crystallizes within the first 10% of denoising steps rather than gradually, yet reward guidance applied across all timesteps outperforms early- or late-only scheduling; and gradient normalization consistently hurts physical metrics, suggesting the magnitude of the reward signal carries meaningful constraint information that must be preserved. These results establish that a lightweight, contrastively trained reward model can meaningfully substitute for expensive physics simulation in diffusion guidance. Future work should investigate analytical replacements for the learned reward using differentiable tools such as RNEA or Pinocchio, adaptive guidance schedules tied to $\hat{x}_0$ stability, and extensions to multi-person and object-interaction settings.

10. Thought-Experiment on Compute

Training used 4 $\times$ A6000 ADA GPUs with PyTorch DDP, `torch.compile`, and AMP (fp16 forward, fp32 gradients), completing 10 epochs on 800 samples in under 30 seconds. With a \$1,000 GPU budget, we would extend this with a minimal PhysDiff-style setup: train lightweight RL policies at a sparse subset of denoising steps, projecting $\hat{x}_0^{(t)}$ to physically plausible motions via simulator imitation with rewards penalizing foot sliding, ground penetration, and floating.

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